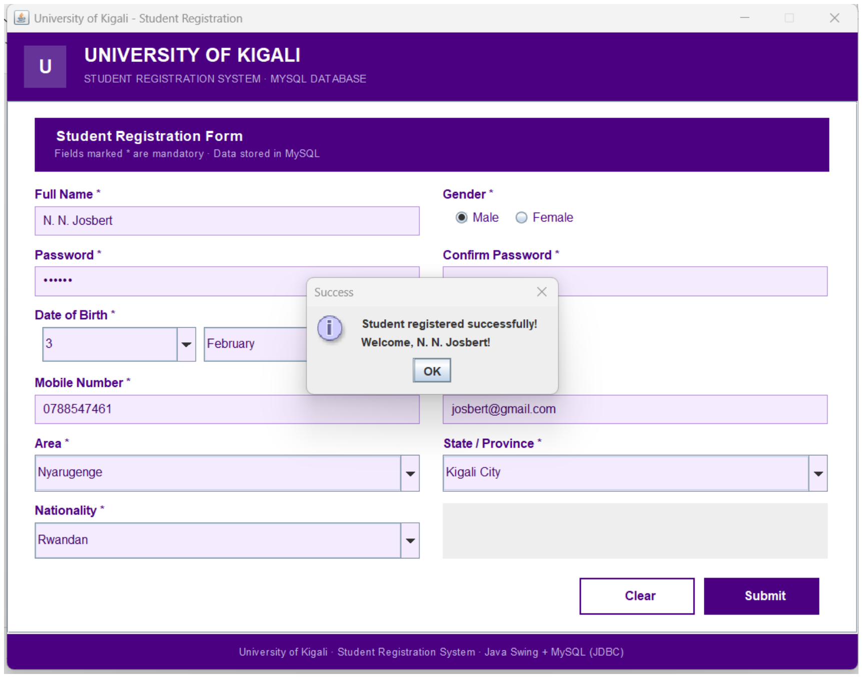
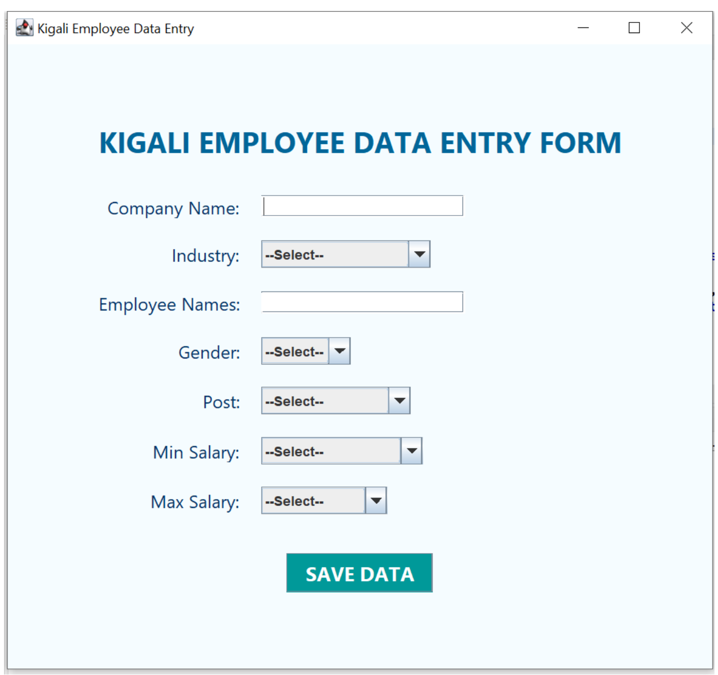
ADVANCED JAVA PROGRAMMING: KEY POINTS

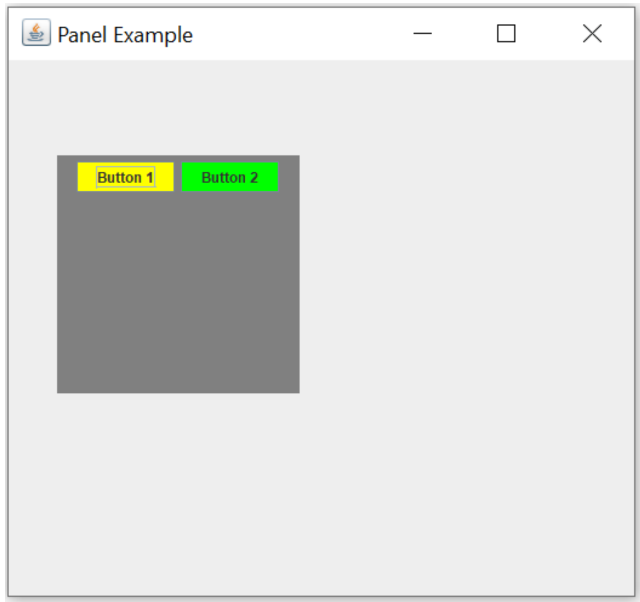
1. Key point 1: JDBC



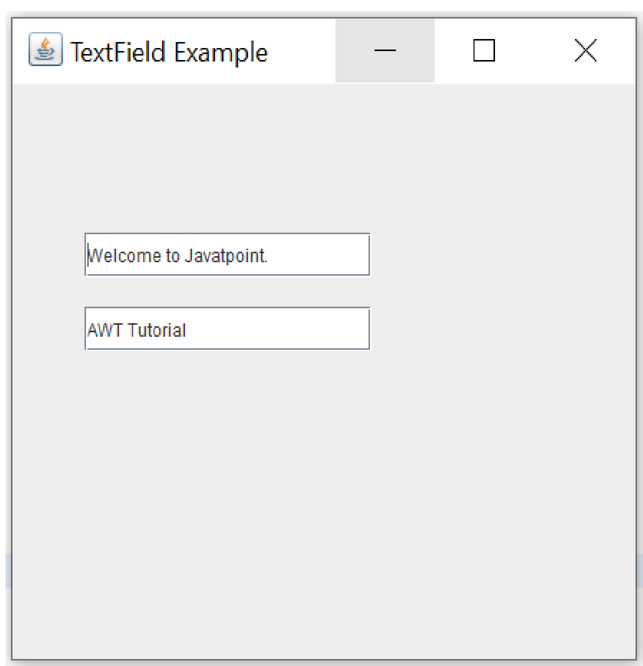
2. Key point 2: JDBC



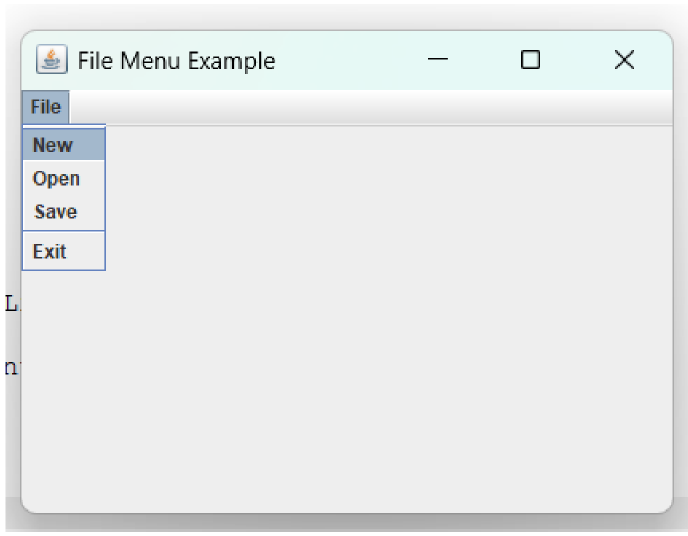
3. **Key point 3: GUI**



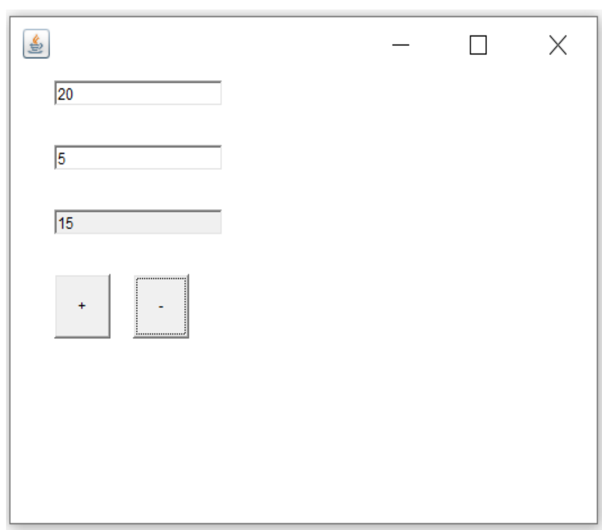
4. **Key point 4: GUI**



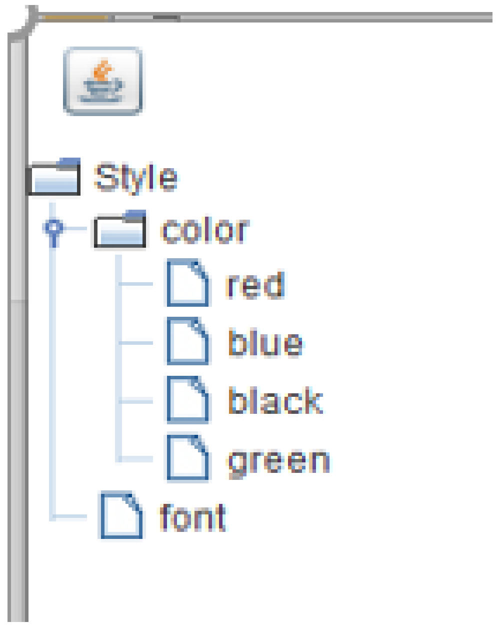
5. **Key point 5: GUI**



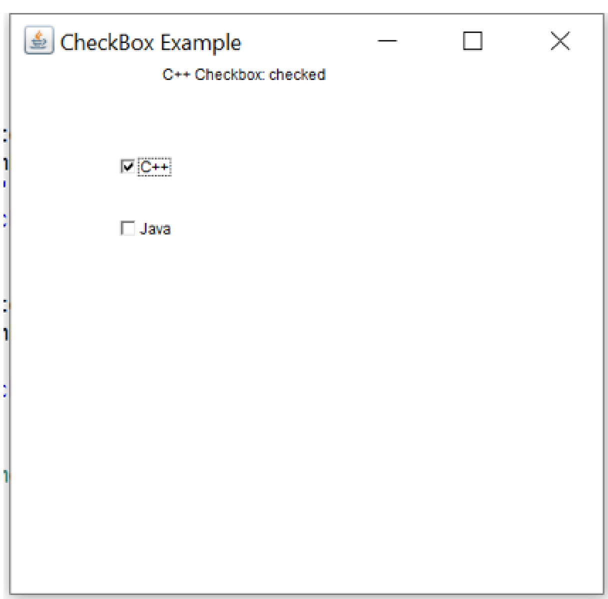
6. **Key point 6: GUI**

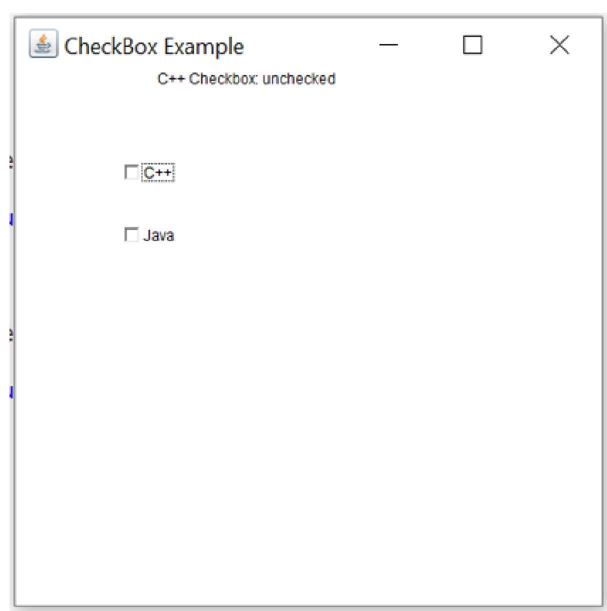


7. **Key point 7: GUI**

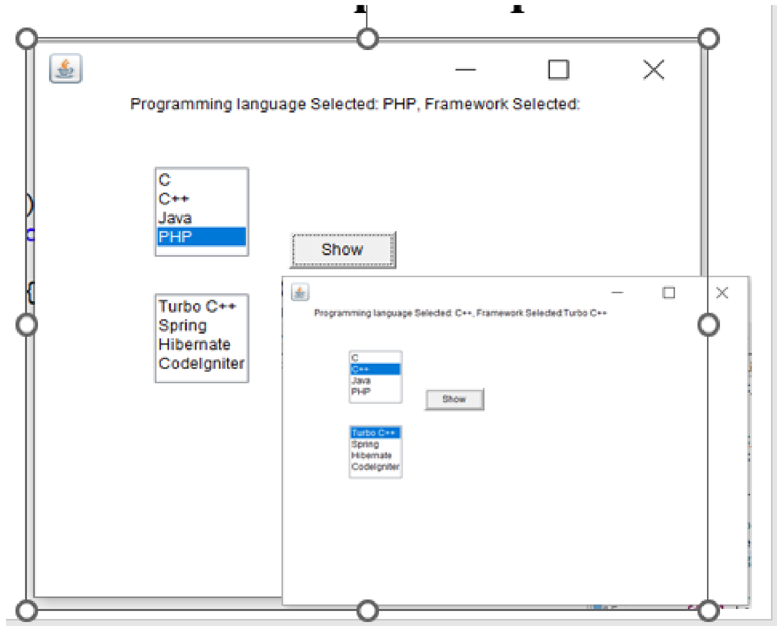


8. **Key point 8: GUI**

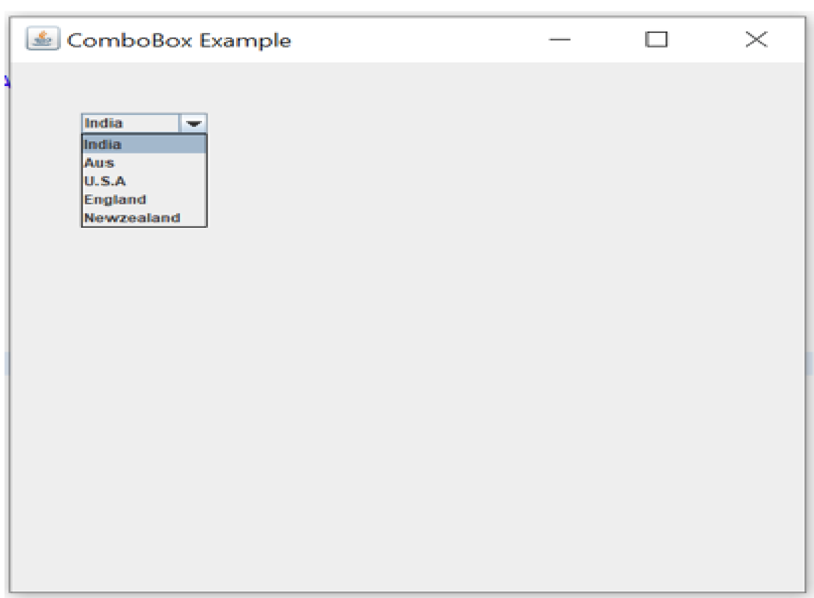




9. **Key point 9: GUI**



10. **Key point 10: GUI**



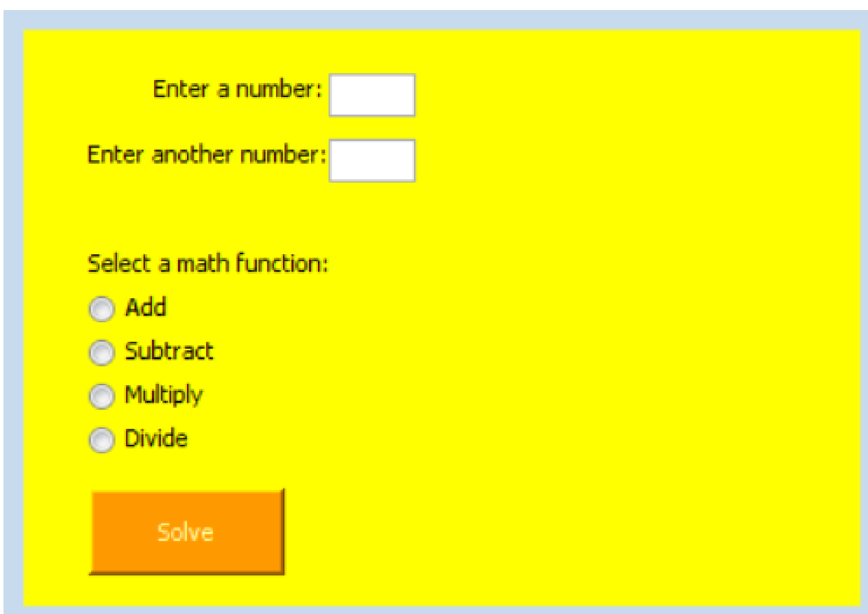
11. Key point 11: Understand Theory:

Examine how the Java compiler, bytecode, and Java Virtual Machine (JVM) work together to enable platform independence in Java applications. Offer a detailed description showing how these components allow a program written once to run on different operating systems without modification, and analyze how this mechanism improves portability and reliability when compared with traditional machine dependent programming languages.

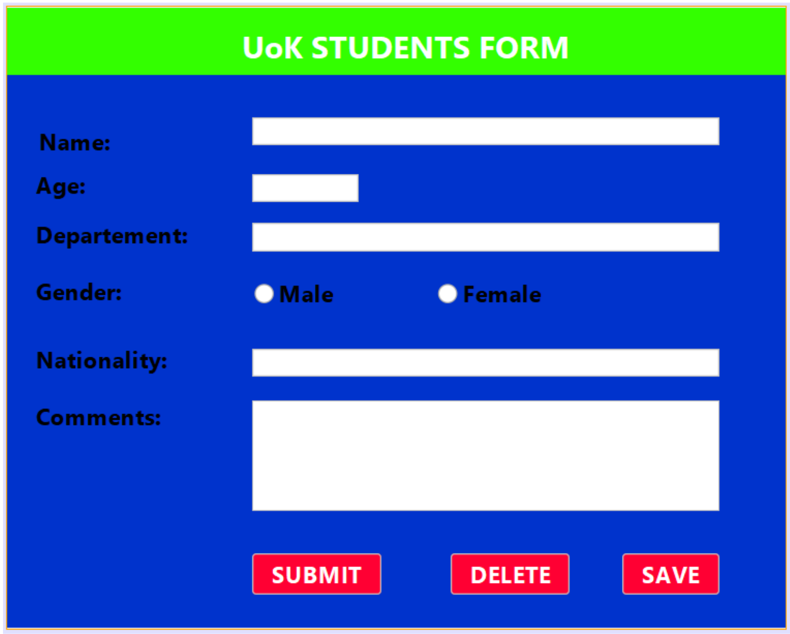
12 . Key point 12: Understand Theory:

Assess the appropriateness of the Java execution environment for developing large scale systems. In your discussion, consider factors such as system performance, security, cross platform compatibility, and ease of maintenance. Explain the advantages of using Java for such a system and also discuss potential limitations or challenges that developers might encounter when implementing Java in a large distributed environment.

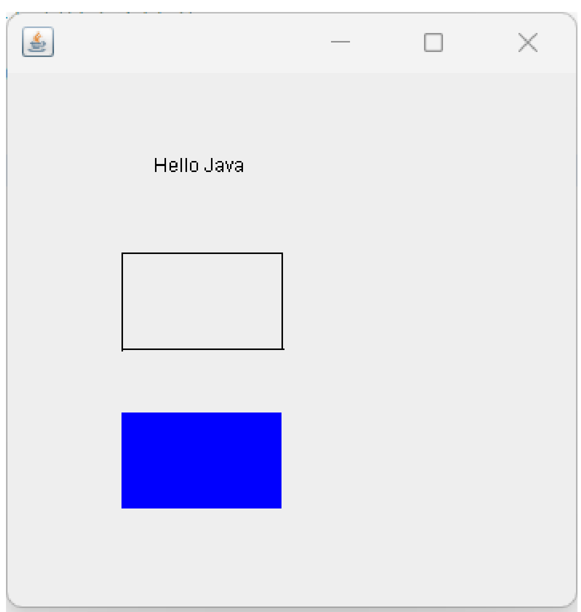
13. Key point 13: NetBeans FORM GUI



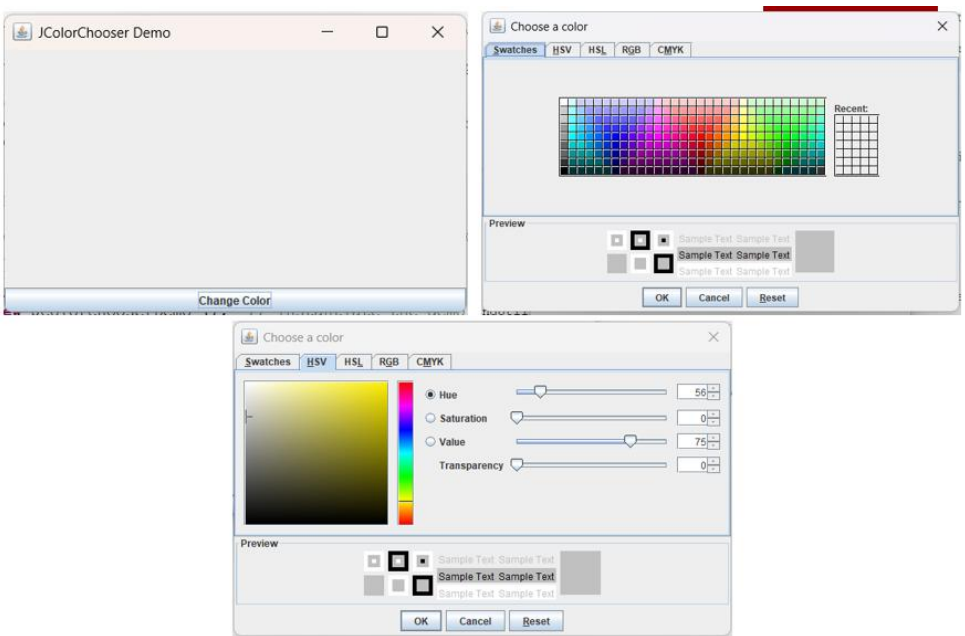
14. Key point 14: NetBeans FORM GUI



15. Key point 15: 2D Shapes and colors



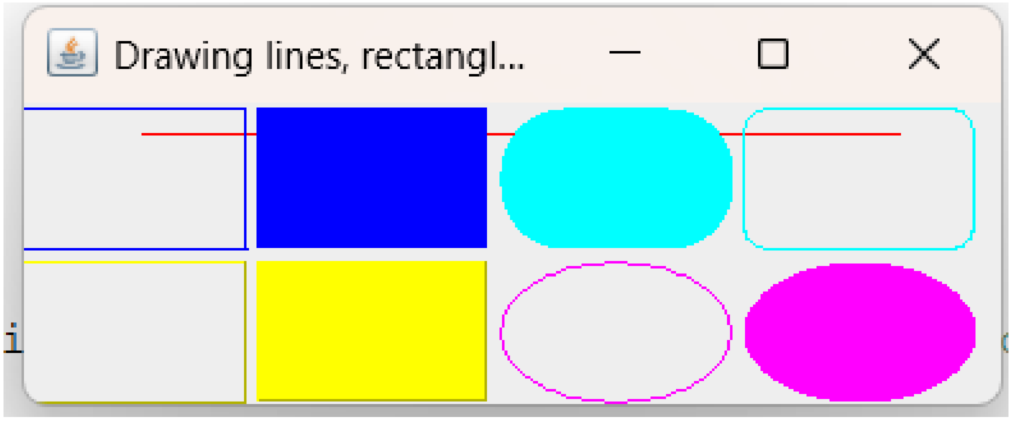
16. Key point 16: 2D Shapes and colors



17. Key point 17: Colors

Color constant	Color	RGB value
public final static Color ORANGE	orange	255, 200, 0
public final static Color PINK	pink	255, 175, 175
public final static Color CYAN	cyan	0, 255, 255
public final static Color MAGENTA	magenta	255, 0, 255
public final static Color YELLOW	yellow	255, 255, 0
public final static Color BLACK	black	0, 0, 0
public final static Color WHITE	white	255, 255, 255
public final static Color GRAY	gray	128, 128, 128
public final static Color LIGHT_GRAY	light gray	192, 192, 192
public final static Color DARK_GRAY	dark gray	64, 64, 64
public final static Color RED	red	255, 0, 0
public final static Color GREEN	green	0, 255, 0
public final static Color BLUE	blue	0, 0, 255

18. Key point 18: 2D Shapes and colors



19. Key point 19: JavaBeans

Write a Java class named Student that follows JavaBeans conventions. The class should include a private property called name to store the student's name. It should provide a no-argument constructor, along with methods that allow setting and retrieving the value of the name property. The class must also support object serialization so that its objects can be saved to a file.

20. Key point 20: JavaBeans

by adding a main application class named MainApp that demonstrates how a Student object can be saved to a file and later loaded back into the program. The program should first create a Student object and assign a name to it. It should then save the object into a file using object serialization. After saving the object, the program should read the same object back from the file using deserialization and display the student's name on the screen. During execution, the program must print confirmation messages. When the object is successfully saved, it should display "Object has been serialized (saved)." When the object is successfully loaded, it should display "Object has been deserialized (loaded)." Finally, it should display the student's name in the format "Student Name: Mugabo Mohammed".

21. Theory related to JavaBeans and BeanInfo.

22. Key Points 22: GUI